

LZW compression

Short for Lempel-Ziv-Welch, this is a popular file compression method used in image compression



Claude Shannon

Laid the foundation of the most profound thesis of all times on data – information theory

How stuff works

Gains from loss

Mohamed Rameez

readersletters@thinkdigit.com

While most high quality movies five years ago could fit into a couple of VCDs, today's demands of high quality prints require high density media such as Blu-ray disks, capable of storing scores of gigabytes.

Compression has indeed been a boon to the media world. You use compression in various ways almost every day. You would've noticed how a JPEG image of the same quality as a bitmap picture is just a fraction in size. Other methods of file compression such as ZIP or RAR are also popular.

Codec

At the heart of a compression tool such as an MP3 ripper is a component known as a codec – short for coder-decoder. A coder is a program / algorithm that is capable of accepting uncompressed raw data (such as a bitmap picture or a wave file) as input and giving a compressed low size file (such as a JPEG image or an MP3 file) as the output. Codecs are often patented technologies, such as the MP3 codec. Sometimes, the same type of encoding (AAC for example) is possible through multiple types of codecs (some open source, some closed).

The compression ratio is the ratio of the file size of the original file to the final compressed file. Compression methods and their associated codecs are largely classified into two – lossy and lossless.

Lossless compression

If the original uncompressed file can be regenerated from the compressed file by the codec, the compression method is considered lossless. In its simplest form, this is often achieved through algorithms that search for repeating patterns in the data and

then replace them with references, making the file smaller. More advanced methods create statistical distributions of the various patterns available, with the highest probability patterns giving maximum compression. These types of compression are often used to store data, for example, ZIP and RAR, and in certain important media applications. Some formats that use lossless compression are PNG, GIF and ATRAC.

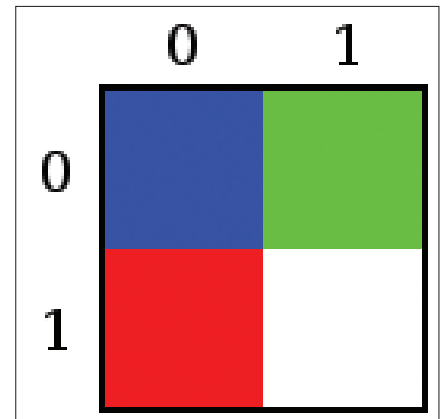
Lossy compression

Lossy compression, is popularly used in media files such as images, audio and video. This type of compression have been designed after a fair amount of study into human sensing abilities. The final product might look and sound the same, but some data is irrevocably lost in the compression process and the codec cannot generate the original data.

JPEG

A common form of lossy compression used in image files, especially on the internet, is the joint photographic experts group (JPEG) and is used in various file formats (file types with different file structures). The most popular among them is JPEG /Exif, used in digital cameras. This allows for

An image encoded in JPEG using successively higher compression ratios from left to right. Notice the successive reduction in quality



A simple 2 pixel by 2 pixel BMP image in 24-bit colour

the inclusion of information such as shooting time, date and exposure time in the image. Another common format, is JFIF, utilised mainly on web pages.

As we know, an image consists of a large number of small pixels. In the most rudimentary file type for an image, known as a bitmap picture, each pixel is denoted by a block of data varying in size from 8 to 32 bits. In an 8-bit BMP, a pixel can take any of 256 colours, while a 32-bit pixel can take 2^{32} colours. The number of pixels in an image increases with higher resolutions. JPEG compression utilises statistical patterns found in nearby pixels so that a region in an image can be

generated using much less data. As such, JPEG is a compression format better suited for when nearby pixels are similar, like a photograph with soft gradients. In images such as drawings with sharper gradients, JPEG leads to observable loss in quality.

The JPEG file structure

A JPEG file consists of many small building blocks known as segments. Each segment can be different in size and characterises a small area on the image consisting of similar pixels. Some segments do not stand for any portion of the